1-Information Technology, the Internet, and You

Introduction

- Purpose
  - Help users become highly efficient and effective computer users
  - How to use:
    1. Apps and application software
    2. Computer hardware
       - Mobile devices
       - Smartphones
       - Tablets
       - Laptops
    3. The Internet
  - Impact of technology on privacy and the environment
  - Role of personal and organizational ethics

the parts of an information system: **people, procedures, software, hardware, data, and the Internet.**

People

- Most important part of any system
- Ways this text helps you become a more efficient and effective computer users
  - Making IT Work for You
  - Tips
  - Privacy
  - Environments
  - Ethics
  - Careers in IT

Software

- Software/Programs
  - Tell the computer how to process data into the form you want
- There are two major kinds of software:
  - System Software
    - Software used by computers
  - Application Software
    - Software you use

System Software

- Enables application software to interact with the computer hardware
- Background software helps manage resources
- Collection of system programs
  - Operating Systems
  - Utilities
  - Device Drivers

System Software cont.

- Operating System
  - Coordinates computer resources
  - Provides the user interface
  - Runs applications
- Embedded operating system
  - Used by Smartphones and tablets
    - Real-time operating systems (RTOS)
- Standalone operating system
  - Used by desktops
- Networking operating systems
  - Used to run networks

System Software Continued

- Utilities
  - Perform specific tasks related to managing computer resources
  - Antivirus Program
    - Protects from viruses
      - Can damage your software or hardware
      - Comprise the security and privacy of personal data

Application Software

- End-user software
- Types of application software
  1. General-Purpose applications
     - Widely used programs
       - Browsers
       - Word Processor
  2. Specialized applications
     - More narrowly focused
       - Web Authoring
  3. Apps
     - Designed for mobile devices
       - Social media apps

Hardware – Types of Computers

- Supercomputers
  - Most powerful computers
- Mainframe computers
  - Process large amounts of data
- Midrange computers
  - Servers
- Personal computers
  - PCs
  - Five types of PCs

Personal Computer Types

Desktop
Laptop (Notebook)
Tablet

Smartphones
Wearables

Personal Computer Hardware

Four basic categories of equipment

- System Unit
- Input/Output
- Secondary Storage
- Communications

System Unit

- System Unit
  - Houses most of the electronic components
- Two important components
  - Microprocessor
  - Memory
    - Holds data currently being processed
    - Holds the processed information before it is output
    - Temporary storage, contents are lost when power is off

Input/Output Devices

Input

Translate data into computer language

Keyboard and Mouse

Output

Translate computer data into usable information

Display and Printer

Secondary Storage

- Holds data and programs even if power is off
- Hard disk
- Solid-state storage
  - No moving parts
  - More reliable
  - Requires less power
- Optical disc
  - Laser technology
  - CDs, DVDs, Blu-ray

Communications

Communication devices

Provide the ability for personal computers to communicate

Modems

Modify audio, video and other types of data for Internet usage

Data

- Raw, unprocessed facts
- Processed data becomes information
- Digital data is stored electronically in files
- Four common types of files
  - Document
  - Worksheet
  - Database
  - Presentation

Files

Document

Worksheet

Database

Presentation

Connectivity and the Mobile Internet

- Connectivity
  - Sharing of information
- Network
  - Communications system connecting two or more devices
  - Central to the concept of connectivity
  - Largest network is the Internet
  - Web provides a multimedia interface for Internet resources

Forces of Technology

- Three things driving the forces of technology
  1. Cloud computing
     - Computers on the Internet
       - Access to more resources
  2. Wireless technology
     - Changing the way we communicate
       - Tablets, smartphones, wearable devices
  3. The Internet of Things (IoT)
     - Continuing development of the Internet
     - Allowing all types of devices to communicate

Careers in IT

- Webmaster
  - Develops and maintains websites and web resources
- Software Engineer
  - Analyzes users' needs and creates application software
- Computer Support Specialist
  - Provides technical support to customers and other users

Technical Writer

Prepares instruction manuals, technical reports, and other scientific or technical documents

Network Administrator

Creates and maintains computer networks

A Look to the Future

- Using and Understanding Information Technology
  - The Internet and the Web
  - Powerful Software
  - Powerful Hardware
  - Security, Privacy and Ethics
  - Organizations
  - Changing Times